

TOPIC 1: Software Project Management

1. The Management Spectrum (The Four P's)

Fresh Detailed Notes: Effective software project management focuses intensely on the "Four P's". A manager who ignores these elements jeopardizes the success of the project.

•**People:** This is the most important element of a successful project. The goal is the cultivation of a motivated, highly skilled software workforce. The Software Engineering Institute (SEI) developed the **People-CMM (Capability Maturity Model)** to help organizations continuously attract, develop, motivate, organize, and retain the workforce to accomplish business objectives. Stakeholders must be properly organized, typically into constituencies: senior managers, technical managers, practitioners, customers, and end-users. *Constantine's Organizational Paradigms* (Closed, Random, Open, Synchronous) are often used to define team structures.

•**Product:** Before planning can happen, the product objectives and scope must be established, alternative solutions identified, and technical/management constraints defined.

•*Scope* defines the primary data, functions, and behaviors. Without it, you cannot estimate cost, assess risk, or schedule realistically.

•*Problem Decomposition:* The scope is decomposed into constituent functions, user-visible data objects, or problem classes until they are easily manageable.

•**Process:** The software process provides the framework from which a comprehensive plan is established. It must be selected appropriately for the people and product. It includes task sets (software engineering tasks, work products, quality assurance points, milestones) and umbrella activities (like configuration management and measurement) that occur throughout the project.

•**Project:** We conduct planned and controlled projects because it is the only known way to manage complexity. To avoid project failure, managers must understand warning signs and adopt a common-sense approach: start on the right foot, maintain momentum, track progress, make smart decisions, and conduct postmortem analyses.

Definitions:

•**People-CMM:** A capability maturity model specifying how organizations should improve their ability to manage and develop their workforce.

•**Project Scope:** A narrative description or set of use-cases describing the functions, features, data, performance, and constraints delivered to end-users.

Exam Exemplar Questions:

•*Question:* "Describe the organizational paradigms suggested by Constantine for structuring a software development team." (7 Marks)

•*Question:* "Analyze the Library Management System... Identify the principal viewpoints which might be considered and organize these..." (6 Marks)

Exam Guide Notes (What to Do & What Not to Do):

•**Do:** When asked about the Management Spectrum, always list all four P's and write at least two lines explaining the significance of each. Emphasize that "People" is the most critical factor.

•**Do:** Memorize Constantine's 4 team paradigms (Closed, Random, Open, Synchronous) as they are the most frequently asked theoretical question under the "People" umbrella.

•**What Not to Do:** Do not confuse "Process" and "Project". Process is the underlying framework and task sets; Project is the actual execution and control of the work.

2. The W5HH Principle

Fresh Detailed Notes: Proposed by Barry Boehm, the W5HH principle is an excellent planning outline used to get to the "essence" of a project, regardless of its size or complexity. It helps define key project characteristics through a specific series of questions:

1.**Why** is the system being developed? (*Identifies the business purpose*)

2.**What** will be done? (*Defines the tasks and deliverables*)

3.**When** will it be accomplished? (*Establishes the project schedule*)

4.**Who** is responsible? (*Defines roles and responsibilities*)

5.**Where** are they organizationally located? (*Identifies team location and communication*)

6. **How** will the job be done technically and managerially? (*Defines the management strategy and technical process*)

7. **How much** of each resource is needed? (*Leads to estimation of people, software, tools*).

Definitions:

• **W5HH:** An acronym standing for Why, What, When, Who, Where, How, and How much; a universal project planning framework.

Exam Exemplar Questions:

• *Question:* "Analyze the series of questions that lead to definition of key project characteristics and the resultant project plan as proposed by Barry Boehm." (6 Marks)

Exam Guide Notes (What to Do & What Not to Do):

• **Do:** Write out the exact W5HH questions sequentially. Provide a brief 3-4 word explanation next to each indicating what project aspect it solves (e.g., "When = Scheduling").

• **What Not to Do:** Do not just write the acronym letters. You must state the full questions to get full marks.

3. Critical Practices

Fresh Detailed Notes: The Airlie Council developed a list of "critical software practices" that are consistently used by highly successful software organizations to ensure project integrity and better-than-average performance. These practices include:

1. **Formal risk management:** Proactively identifying and mitigating risks.

2. **Empirical cost and schedule estimation:** Using data-driven models to predict project resources.

3. **Metrics-based project management:** Tracking progress using objective, quantitative measures.

4. **Earned value tracking:** Quantitatively assessing the "percent of completeness" of a project.

5. **Defect tracking against quality targets:** Monitoring errors to ensure the product meets quality standards.

6. **People-aware project management:** Cultivating and organizing the team effectively.

Definitions:

• **Earned Value Analysis (EVA):** A measure of progress that enables managers to assess the percent of completeness of a project using quantitative analysis rather than gut feeling.

Exam Exemplar Questions:

• *Question:* Describe the critical software practices for performance-based management.

Exam Guide Notes (What to Do & What Not to Do):

• **Do:** Use this list if asked about "best practices" or "performance-based management practices" for software projects.

TOPIC 2: Estimation for Software Projects

1. Software Project Estimation & Decomposition Techniques

Fresh Detailed Notes: Estimation of resources, cost, and schedule requires experience, access to historical metrics, and the courage to commit to quantitative predictions. Because estimation carries inherent uncertainty, you should never rely on just one technique; at least two different methods should be used and reconciled.

To estimate accurately, planners must decompose (break down) the problem into smaller, manageable functions or objects. The two primary conventional decomposition techniques use lines of code (LOC) or function points (FP) as the measurement metric.

A. LOC (Lines of Code) Method Computes project cost and effort using the estimated lines of code and historical organizational data.

• **Numerical Formulas:**

1. **Expected LOC (S):** Calculated using optimistic (S_{opt}), most likely (S_m), and pessimistic (S_{pess}) estimates: $S = (S_{opt} + 4 * S_m + S_{pess}) / 6$
2. **Cost per LOC:** Labor rate per month / Organization's Average Productivity (OAP)
3. **Total Cost:** Total LOC * Cost per LOC
4. **Estimated Effort (Person-Months):** Total LOC / OAP

B. FP (Function Point) Method Estimates the functionality delivered by the software based on the Information Domain Values (External inputs, outputs, inquiries, internal logical files, external interface files).

• **Numerical Formulas:**

1. **Value Adjustment Factor (VAF):** $0.65 + (0.01 * \Sigma Fi)$ (where ΣFi is the sum of complexity adjustment questions).
2. **Estimated FP (Total FP):** Count Total * VAF.
3. **Cost per FP:** Labor rate per month / OAP.
4. **Total Cost:** Total FP * Cost per FP.
5. **Estimated Effort (Person-Months):** Total FP / OAP.

C. Use-Case Method Similar to the LOC method, but the "volume" of the software is measured in the number of Use-Cases. Effort and cost are extrapolated using historical productivity data based on use-case counts.

Symbol Definitions for Formulas:

- **OAP:** Organization's Average Productivity (measured in LOC/month or FP/month).
- **ΣFi / $\Sigma(f_i)$:** Sum of complexity factors (Information Domain Values).
- **VAF:** Value Adjustment Factor, scaling the raw count based on complexity.

Exam Exemplar Questions:

- **LOC Problem:** "An organizations average productivity is 550 LOC/pm. The average labor rate is \$9750 per month. Calculate (i) Cost per line of Code (ii) Overall Project cost if the proposed project has 282000 LOC (iii) Estimated effort in person-months." (6 Marks)
- **FP Problem:** "If an organizations productivity is 12 FP/pm based on a burdened labor rate of \$20000 per month. The Count total 450 and $\Sigma(f_i)$ is 58. Calculate: a. Cost per FP b. Estimated effort in person-months c. Estimated project cost" (6 Marks)

Exam Guide Notes (What to Do & What Not to Do):

- **Do:** Write down the formula for every single step explicitly before plugging in the numbers. Even if your final math is slightly off, examiners award partial marks for the correct formulas. Include units (e.g., "\$", "person-months").
- **What Not to Do:** In FP problems, **NEVER multiply the "Count Total" directly with the Cost per FP.** This is the most common student mistake. You **MUST** first calculate the VAF, then multiply **Count Total * VAF** to find the actual *Estimated Total FP* before computing costs.

2. Empirical Estimation Models

Fresh Detailed Notes: Empirical estimation models are statistical models derived from historical data of past software projects. They provide a formulaic way to predict effort.

- **General Form:** $Effort = \text{tuning coefficient} \times (\text{size})^{\text{exponent}}$ (usually derived as person-months of effort).
- **The Software Equation:** A dynamic multivariable empirical model.
 - Formula: $E = [LOC \times B^{0.333/P^3}] \times [1/t^4]$
 - Where: E = effort in person-months or years; t = project duration; B = special skills factor; P = productivity parameter.
- **COCOMO II (Constructive Cost Model):** The industry-standard empirical model. It addresses different stages of the software process using three sub-models:
 - *Application composition model:* Used during the early stages (UI prototyping, technology evaluation).

- Early design stage model*: Used once requirements are stabilized and basic software architecture has been established.
- Post-architecture-stage model*: Used during the actual construction of the software.

Definitions:

- Empirical Model**: An estimation model based on observation and historical data rather than pure mathematical theory.

Exam Exemplar Questions:

- Question*: "Explain the empirical estimation models. Detail the general form and describe the sub-models of COCOMO II."

Exam Guide Notes (What to Do & What Not to Do):

- Do**: Memorize the general empirical formula **Effort = tuning coefficient * size^{exponent}** and the three specific stages of COCOMO II. State clearly that these models rely on historical data.